Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Software engineer with 2+ years building **high-performance systems**, **ML algorithms**, and **data pipelines**, with strong foundations in algorithm design, software optimization, and applied machine learning in research and production environments.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Data Structures, Linear Algebra, Statistics

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **high-performance platform services** for 500+ users, reducing request latency from **1.2s to 350ms (71%)** by profiling execution paths, optimizing algorithms, and tuning database query patterns for mission-critical analytics.
- Implemented **automated validation and data pipelines** for 5,000+ monthly jobs, applying statistical analysis techniques to detect regressions, data-quality drift, and runtime anomalies in production environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **concurrent batch-processing engine** handling 10,000+ daily requests, implementing efficient algorithms for task scheduling and resource allocation in high-throughput production workloads.
- Built **ML-powered classification workflows** with multimodal models and OCR, achieving 90%+ accuracy while prototyping solutions in an agile development environment and iterating based on analysis results.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **GPT-based Decision Transformer** for autonomous driving using PyTorch, processing 100,000+ trajectory records and improving long-horizon planning by **18%** through spatial-temporal encoding and multi-head attention.
- Designed **experimentation pipelines and simulation environments** for algorithm evaluation, using high-fidelity modeling to quantify performance benefits and validate technical constraints across closed-loop scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API microservices and ETL pipelines** using AWS, improving response time by **30%** through algorithm optimization, caching strategies, and software analysis and profiling techniques.
- Maintained **20+ automated test suites** using Jenkins, PyTest, and Mockito, increasing reliability by **68%** and applying rigorous software testing and optimization practices across the development lifecycle.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **real-time data synchronization system** for 2M+ records using WebSockets, optimizing throughput and reducing sync conflicts by **33%** through efficient state management and algorithm design.
- Automated **Docker-based build and deployment pipelines** with Jenkins, cutting release time by **50%** and working with all aspects of the software development lifecycle from design to deployment.

Skills

Languages: **C++**, **Python**, Java, SQL, JavaScript/TypeScript, Bash
ML & Algorithms: **PyTorch**, Transformers, supervised/unsupervised learning, classification, optimization
Databases: **PostgreSQL**, MySQL, **MongoDB**, NoSQL, query optimization, data pipelines
Systems & Tools: **Linux**, Docker, Git, CI/CD, Jenkins, AWS, algorithm profiling
Mathematics: Linear algebra, statistics, optimization, algorithm analysis, numerical methods

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, OS Internals*

- Built a **scheduling algorithm simulator** implementing FCFS, SJF, SRT, RR, and MLQ, reducing average turnaround time by **25%** through algorithm optimization across 1K+ simulated process combinations.

CaptionCraft Live | *PyTorch, Transformers, Flask, Deep Learning*

- Benchmarked **transformer architectures** (BLIP, GIT, ViT-GPT-2) on 10K images / 100K captions, improving caption accuracy from 81% to **96%** through model analysis, hyperparameter tuning, and inference optimization.

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Built a **containerized REST processing API** on AWS CodePipeline and ECS/Fargate with automated validation gates, achieving 87% accuracy and production-grade reliability for data processing workflows.